

PULSE DIAGNOSIS SYSTEM - COMPONENTS SPECIFICATIONS & PINOUT GUIDE

Complete Hardware Reference for ECG,
PPG, h-FABP Optical Sensor System

Date: December 2025

Status: Complete Component List with Pinouts and Specifications

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COMPONENT LIST & COST

Component	Part Number	Quantity	Cost (₹)	Supplier
Microcontroller				

Component	Part Number	Quantity	Cost (₹)	Supplier
ESP32 DevKit V1	ESP32-DEVKITC-32U	1	450	Amazon.in
Biosensors				
AD8232 ECG Module	AD8232 Breakout	1	550	Amazon.in
MAX30102 PPG Module	MAX30102 Breakout	1	250	Amazon.in
UV LED 365nm (5mm)	365nm UV LED	3	60	Local/GeeksforGeeks
BPW34 Photodiode	BPW34	2	80	VCC Online
Communication				
SIM800L GSM Module	SIM800L Breakout	1	450	Amazon.in
Power Management				
TP4056 Li-Po Charger	TP4056 Module	1	100	Amazon.in
18650 Battery 1000mAh	18650 3.7V	2	300	Amazon.in
Display & Interface				
OLED 0.96" I2C Display	SSD1306 0.96"	1	200	Amazon.in
Passive Components				
Resistor Kit (10Ω–1MΩ)	Assorted 1/4W	1	50	Local
Capacitor Kit (100nF–100μF)	Ceramic + Electrolytic	1	50	Local
Connecting Wires	22AWG Jumper Wires	1 pack	100	Amazon.in
Breadboard/PCB	Prototype Board	1	150	Local
Miscellaneous				
Medical-Grade Silicone Patch	Eco-flex Silicone	1 sheet	80	Amazon.in
2N7000 MOSFET	2N7000 N-Channel	5	15	Local
Push Buttons (tactile)	6x6mm buttons	10	20	Local
LED Indicators (various colors)	5mm LEDs	20	30	Local
Total Estimated Cost			₹2,845	

ESP32 DEVKIT V1 – PINOUT & SPECIFICATIONS

Physical Specifications

- **Dimensions:** 49 × 26 × 13 mm
- **Weight:** ~10 grams
- **Operating Voltage:** 3.3V (USB input 5V with onboard regulator)
- **Flash Memory:** 4 MB

- **RAM:** 520 KB (SRAM) + 4 MB (PSRAM on some variants)

GPIO Pinout (Left side, from top to bottom)

Pin #	GPIO	Function	Type	Notes
1	GND	Ground	-	Power reference
2	23	GPIO23	I/O	General purpose
3	22	GPIO22	I/O	I2C SCL (recommended)
4	TX0	GPIO1	Serial	UART0 TX (Programming)
5	RX0	GPIO3	Serial	UART0 RX (Programming)
6	21	GPIO21	I/O	I2C SDA (recommended)
7	GND	Ground	-	Power reference
8	19	GPIO19	I/O	General purpose
9	18	GPIO18	I/O	SPI CLK (optional)
10	5	GPIO5	I/O	SPI CS (optional)
11	17	GPIO17	I/O	UART2 TX (optional)
12	16	GPIO16	I/O	UART2 RX (optional)
13	4	GPIO4	I/O	General purpose
14	0	GPIO0	I/O	BOOT button (low = bootloader)
15	2	GPIO2	I/O	General purpose
16	15	GPIO15	I/O	General purpose

GPIO Pinout (Right side, from top to bottom)

Pin #	GPIO	Function	Type	Notes
1	3V3	3.3V Power	Power	Regulated output (~500mA max)
2	EN	Enable	Reset	Pull-to-ground to reset
3	36	GPIO36	ADC Input	ADC0 - Analog input
4	39	GPIO39	ADC Input	ADC3 - Analog input
5	34	GPIO34	ADC Input	Input-only (no output)
6	35	GPIO35	ADC Input	Input-only (no output)
7	32	GPIO32	I/O	ADC4 - Analog input
8	33	GPIO33	I/O	ADC5 - Analog input
9	25	GPIO25	I/O	DAC (digital-to-analog)
10	26	GPIO26	I/O	DAC (digital-to-analog)
11	27	GPIO27	I/O	General purpose
12	14	GPIO14	I/O	General purpose
13	12	GPIO12	I/O	General purpose
14	13	GPIO13	I/O	General purpose
15	GND	Ground	-	Power reference
16	VIN	USB Input	Power	5V from USB (before regulator)

Recommended Pin Assignments for Your Project

I2C Communication (for OLED, MAX30102):

SDA → GPIO 21 (Pin 6, left side)

SCL → GPIO 22 (Pin 3, left side)

Serial Communication (ECG module - optional analog):

RX → GPIO 16 (UART2, right side pin 5)

TX → GPIO 17 (UART2, right side pin 4)

Analog Inputs (for biosensors):

AD8232 ECG Output → GPIO 32 (ADC4, right side pin 7) or GPIO 33 (ADC5)

h-FABP Photodiode → GPIO 35 (ADC3, right side pin 4)

Digital Inputs/Outputs:

Push Button → GPIO 0 (left side) with pull-up resistor

LED Indicator → GPIO 4 (left side)

Buzzer → GPIO 27 (right side)

Power:

3.3V → VCC pins on all modules

GND → GND pins on all modules

ADC Specifications

- **Resolution:** 12-bit (0–4095 counts)
- **Range:** 0–3.3V
- **Number of Channels:** 8 ADC channels
- **Sampling Rate:** Up to 200 kSPS (kilosamples per second)
- **Recommended for analog sensor input**

AD8232 ECG MODULE – PINOUT & SPECIFICATIONS

Physical Appearance

- Red PCB, approximately 2cm × 2cm
- 5 pin header on one edge
- On-board signal processing amplifier

Pinout Diagram

[AD8232 Module - Top View]

LO-	LO+	OUT	3.3V	GND
1	2	3	4	5

Pin Description

Pin #	Label	Function	Connection
1	LO-	Right Leg Drive Output (Negative)	Patient's right leg (electrode pad)
2	LO+	Right Leg Drive Output (Positive)	Patient's left leg (electrode pad)
3	OUT	Analog ECG Signal Output	→ ESP32 GPIO 32 or 33 (ADC)
4	3.3V	Power Supply Input	→ ESP32 3.3V
5	GND	Ground	→ ESP32 GND

Technical Specifications

- **Supply Voltage:** 3.0–5.5V (optimized for 3.3V)
- **Operating Current:** ~5mA
- **Signal Gain:** 100 V/V (adjustable via on-board resistors)
- **Bandwidth:** 0.5–40 Hz (suitable for ECG signals)
- **Output Range:** 0–3.3V (ratiometric to supply)
- **Max Output Current:** 10mA

How It Works

1. **Three electrodes** (RA: Right Arm, LA: Left Arm, RL: Right Leg) placed on patient
2. ECG electrodes connected to module via conductive pads
3. **LO+ and LO-** drive the right leg to reduce common-mode noise (prevents 60Hz interference)
4. **OUT** provides amplified, filtered ECG waveform
5. Output digitized via ESP32 ADC at 100 Hz sampling rate

ECG Electrode Placement (Standard 3-Lead)

```
LA (Left Arm)
|
[electrode]
|
RA ← [CHEST] → LO (Left Leg)
[electrode]      [electrode]
```

Wiring Example

```
// Arduino Code Example
int ecgPin = 32; // ADC input on ESP32

void setup() {
  analogSetAttenuation(ADC_11db); // Set to read 3.3V
}

void loop() {
  int ecgValue = analogRead(ecgPin); // Read 0-4095
  float ecgVoltage = ecgValue * (3.3 / 4095); // Convert to voltage
  Serial.println(ecgVoltage);
}
```

MAX30102 PPG MODULE – PINOUT & SPECIFICATIONS

Physical Appearance

- Small red PCB with two IR LEDs on front surface
- 5-pin header for connections
- I2C communication interface

Pinout Diagram

[MAX30102 Module - Front View]

GND	SCL	SDA	3.3V	INT
1	2	3	4	5

Pin Description

Pin #	Label	Function	Connection
1	GND	Ground	→ ESP32 GND
2	SCL	I2C Clock Line	→ ESP32 GPIO 22
3	SDA	I2C Data Line	→ ESP32 GPIO 21
4	3.3V	Power Supply	→ ESP32 3.3V
5	INT	Interrupt Output	→ ESP32 GPIO 35 (optional)

Technical Specifications

- **Supply Voltage:** 3.0–5.5V (optimized for 3.3V)
- **Operating Current:**
 - Idle: ~0.7 mA
 - Two LEDs on: ~100 mA (typical)
 - SpO2 measurement: ~11 mA average
- **Sampling Rate:** 100 Hz, 200 Hz, 400 Hz, 800 Hz, 1000 Hz, 1600 Hz, 3200 Hz (configurable)
- **LED Wavelengths:**
 - Red: 660 nm (850 mW/m²)
 - IR: 880 nm (850 mW/m²)
- **Measurement Range:** 0–100% SpO2
- **I2C Address:** 0x57 (default)
- **FIFO Buffer:** 32 samples (circular)

How PPG Works for Your Project

1. **Red & IR LEDs** flash alternately on skin
2. **Photodiode** measures reflected light
3. **Signal ratio** (Red/IR) indicates blood oxygen saturation (SpO2)
4. **Pulse waveform** visible at each light wavelength
5. **Pulse quality** (amplitude, regularity) extracted from waveform

Heart Rate & SpO2 Detection

The MAX30102 contains:

- **Two LEDs** (Red 660nm + IR 880nm)
- **One photodiode** (measures both wavelengths via time-division multiplexing)
- **24-bit ADC** for high-resolution measurements

Wiring Example

MAX30102		ESP32
GND	→	GND
SCL	→	GPIO 22 (I2C SCL)
SDA	→	GPIO 21 (I2C SDA)
3.3V	→	3.3V
INT	→	GPIO 35 (optional interrupt)

Arduino Code Example

```
#include <Wire.h>
#include "MAX30105.h"

MAX30105 particleSensor;

void setup() {
  Wire.begin(21, 22); // SDA=21, SCL=22
  if (!particleSensor.begin(Wire, I2C_SPEED_FAST)) {
    Serial.println("MAX30102 not found!");
    while(1);
  }

  // Configure sensor
  particleSensor.setup();
  particleSensor.setSampleRate(100); // 100 Hz
}

void loop() {
  uint32_t irValue = particleSensor.getIR();
  uint32_t redValue = particleSensor.getRed();

  Serial.print("IR: "); Serial.print(irValue);
  Serial.print(" | RED: "); Serial.println(redValue);
}
```

SIM800L GSM MODULE – PINOUT & SPECIFICATIONS

Physical Appearance

- Blue PCB with SIM card slot
- 4-pin header for power/ground
- 4-pin header for UART (TX/RX/GND/3.3V)
- Integrated antenna (usually on-board)

Pinout Diagram

[SIM800L Module - Top View]

Power Header				UART Header	
VCC	GND			TX	RX
1	2	3	4	1	2
GND (2)		TX (3)	RX (4)		

Full pinout:

Pin 1: VCC (4.2V input)

Pin 2: GND

Pin 3: TX (3.3V output)

Pin 4: RX (3.3V input)

Pin 5: GND (shared)

Pin 6: 3.3V Power Out

Pin Description

Pin #	Label	Function	Connection
1	VCC	Power Input (4.2V DC)	→ TP4056 output or Li-Po battery +
2	GND	Ground	→ ESP32 GND
3	TX	UART Transmit (3.3V)	→ ESP32 RX (GPIO 16)
4	RX	UART Receive (3.3V)	→ ESP32 TX (GPIO 17) via 1k resistor
5	GND	Ground	→ ESP32 GND
6	3.3V	Power Output	Typically not used

Technical Specifications

- **Supply Voltage:** 3.4–4.2V DC (NOT 3.3V!)
- **Operating Current:**
 - Idle (no signal): ~5 mA
 - Searching for network: ~50 mA
 - Talking (SMS/call): ~100–200 mA
 - Peak: 2A during transmission
- **UART Baud Rate:** 9600 bps (default), configurable to 115200 bps
- **Frequency Bands:** GSM 850/900/1800/1900 MHz
- **SMS Capability:** Text message send/receive
- **Data:** 2G GPRS capable (slow)

⚠ CRITICAL: Power Supply Requirements

SIM800L MUST NOT be powered by ESP32 3.3V directly—it draws too much current and will crash the ESP32!

Recommended Power Supply:

- Use **TP4056 output** (Li-Po battery charging module) directly
- Or dedicated **LDO regulator (5V → 4.2V)** with **large capacitor bank**
- Add **1000µF capacitor** at VCC pin to buffer current spikes

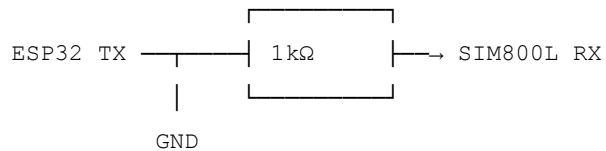
Wiring (CRITICAL)

SIM800L VCC → TP4056 output (or 4.2V source)

SIM800L GND → ESP32 GND (COMMON GROUND)

SIM800L TX → ESP32 GPIO 16 (RX) [direct connection, 3.3V safe]

SIM800L RX → ESP32 GPIO 17 (TX) [via 1k resistor voltage divider]



Arduino Code Example

```
#include <SoftwareSerial.h>

SoftwareSerial SIM800(16, 17); // RX=16, TX=17

void setup() {
  SIM800.begin(9600);
  delay(1000);

  // Send test command
  SIM800.println("AT");
  delay(500);
  if (SIM800.available()) {
    Serial.println(SIM800.readString());
  }
}

void loop() {
  // Send SMS
  SIM800.println("AT+CMGF=1"); // Set text mode
  delay(100);
  SIM800.println("AT+CMGS=\"+91XXXXXXXXXX\""); // Phone number
  delay(100);
  SIM800.println("Heart Rate Alert!");
  delay(100);
  SIM800.write(26); // Ctrl+Z to send
}
```

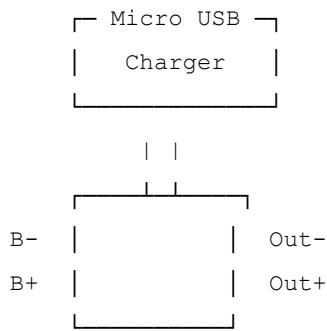
TP4056 LI-PO CHARGER MODULE – PINOUT & SPECIFICATIONS

Physical Appearance

- Blue PCB, ~2cm × 3cm
- Micro USB charging port on top
- Two 2-pin headers: Input (B+ / B-) and Output (Out+ / Out-)

Pinout Diagram

[TP4056 Module - Top View]



Input Header (left):

Pin 1: B- (Battery negative)

Pin 2: B+ (Battery positive)

Output Header (right):

Pin 1: Out- (Output negative)

Pin 2: Out+ (Output positive)

Pin Description

Pin #	Label	Function	Connection
1 (Left)	B-	Battery Negative	→ 18650 Li-Po battery (-) terminal
2 (Left)	B+	Battery Positive	→ 18650 Li-Po battery (+) terminal
1 (Right)	Out-	Output Negative (Discharge)	→ SIM800L GND & ESP32 GND
2 (Right)	Out+	Output Positive	→ SIM800L VCC

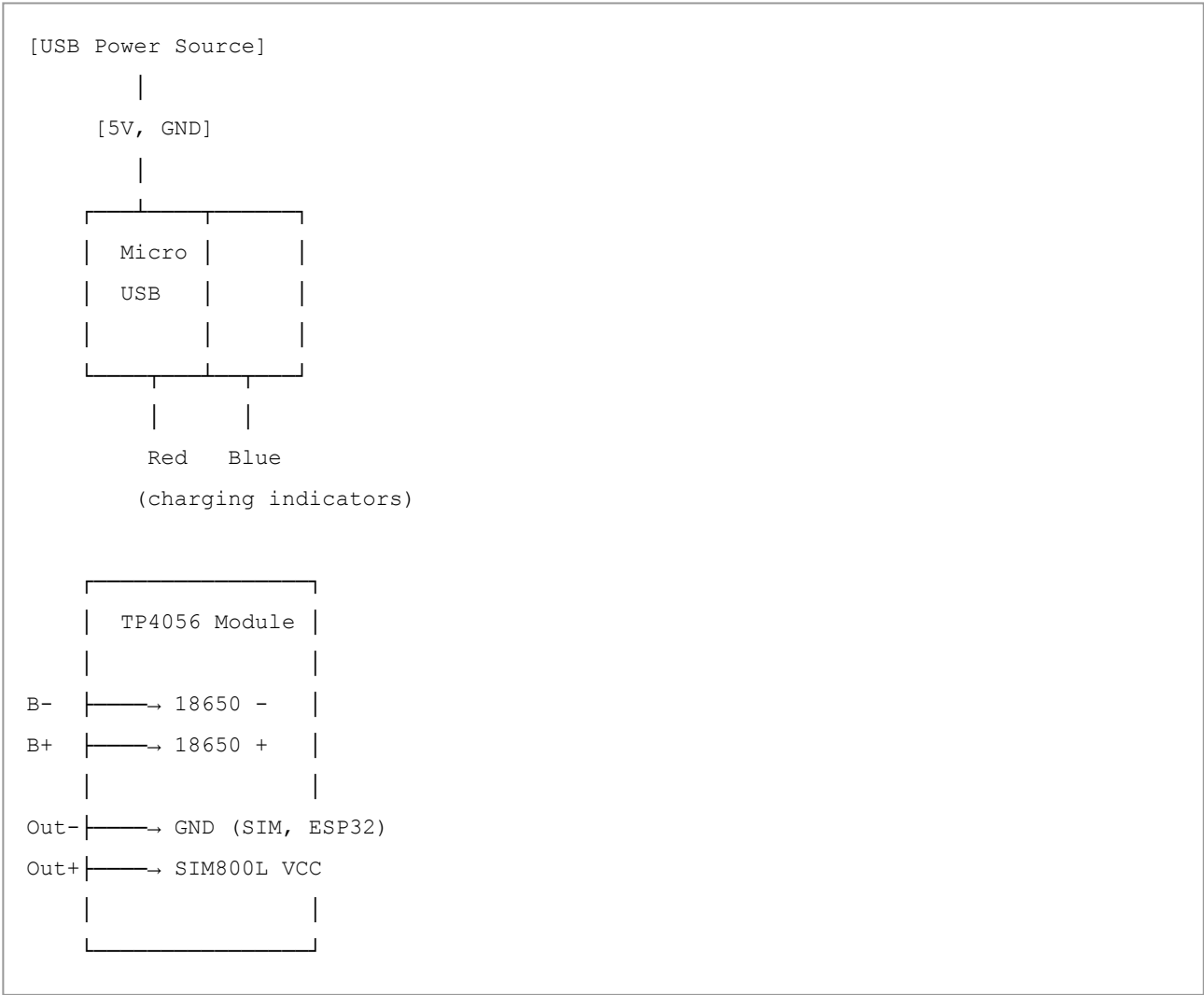
Technical Specifications

- **Input Voltage:** 5V DC via Micro USB
- **Output Voltage:** 3.7–4.2V (Li-Po regulation)
- **Max Charging Current:** 1A (typical), 500mA default
- **Battery Type:** Li-Po/Li-Ion single cell (3.7V nominal)
- **Protection:** Overcharge, over-discharge, short circuit protection
- **Charging Time:** ~2 hours (500mA), ~1 hour (1A charging)
- **Operating Current (no load):** ~1 mA

LED Indicators

- **Red LED:** Charging in progress
- **Blue LED:** Charging complete
- **Both off:** No battery connected or full

Wiring Example



Arduino: Battery Monitoring (Optional)

```
int batteryVoltagePin = 33;  // ADC5 on ESP32

void setup() {
  analogSetAttenuation(ADC_11db);  // Read up to 3.3V
}

void loop() {
  int raw = analogRead(batteryVoltagePin);
  float voltage = raw * (3.3 / 4095) * 2;  // x2 if voltage divider used

  Serial.print("Battery: ");
  Serial.print(voltage);
  Serial.println("V");

  if (voltage < 3.4) {
    Serial.println("LOW BATTERY!");
  }
}
```

UV LED 365nm & BPW34 PHOTODIODE – PINOUT & SPECIFICATIONS

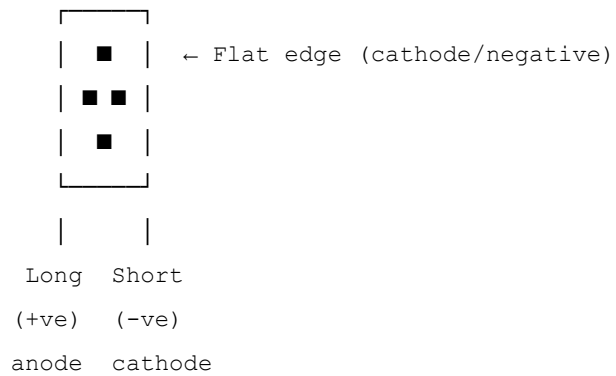
UV LED 365nm (h-FABP Excitation Light Source)

Physical Specifications

- **Type:** 5mm through-hole LED
- **Wavelength:** 365 nm (ultraviolet)
- **Forward Voltage:** 3.3–3.8V
- **Forward Current:** 20 mA (typical), 30 mA (max)

Pinout

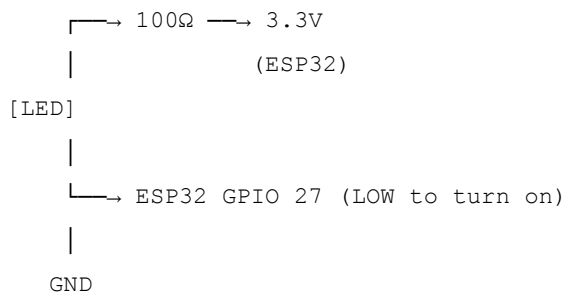
[UV LED - Front View]



Pin Description

Leg	Label	Function	Connection
Long	Anode (+)	Positive terminal	→ 100Ω resistor → 3.3V
Short	Cathode (-)	Negative terminal	→ ESP32 GPIO 27 or via transistor

Wiring



Control Code

```
int uvLedPin = 27;

void setup() {
  pinMode(uvLedPin, OUTPUT);
  digitalWrite(uvLedPin, LOW); // Turn on LED
}

void loop() {
  digitalWrite(uvLedPin, HIGH); // LED OFF
  delay(1000);
  digitalWrite(uvLedPin, LOW); // LED ON
  delay(1000);
}
```

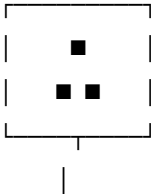
BPW34 Photodiode (h-FABP Detection)

Physical Specifications

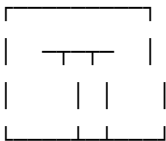
- **Type:** Silicon photodiode, high-speed
- **Peak Sensitivity:** 900 nm (near-infrared), but sensitive to UV/visible
- **Reverse Voltage Rating:** 60V
- **Capacitance:** ~20 pF (important for high-impedance signal)
- **Dark Current:** ~2 nA (very low, good for weak light detection)

Pinout

[BPW34 Photodiode - Top View]



Cathode (K)



← Band on side marks cathode

Anode (A) Cathode (K)

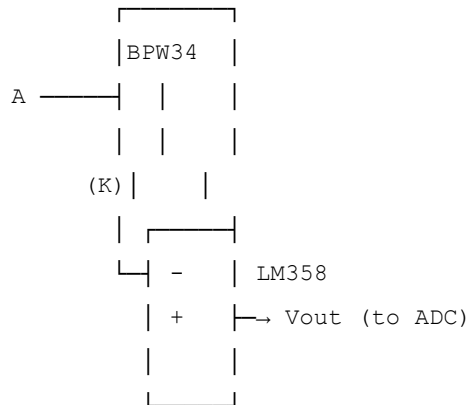
Pin 1 Pin 2

Pin Description

Pin #	Label	Function	Connection
1	Anode (A)	Positive terminal	→ Transimpedance amplifier input
2	Cathode (K)	Negative terminal (bias)	→ -5V to 0V reverse bias

Transimpedance Amplifier Circuit (High-Impedance Signal Conditioning)


```
// For weak UV signals (h-FABP), use a transimpedance amplifier
// (e.g., LM358 op-amp in inverting configuration)
```



Arduino Code with ADC

```
int photodiodePin = 35; // ADC3 on ESP32

void setup() {
  analogSetAttenuation(ADC_11db); // 0-3.3V range
}

void loop() {
  int photoValue = analogRead(photodiodePin);
  float voltage = photoValue * (3.3 / 4095);

  // h-FABP concentration analysis (requires calibration)
  float hFabpConc = voltage * 0.27; // Example: 0.27 ng/mL per volt

  Serial.print("h-FABP: ");
  Serial.print(hFabpConc);
  Serial.println(" ng/mL");
}
```

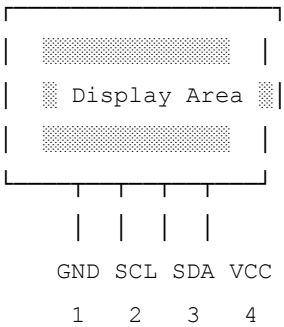
OLED 0.96" I2C DISPLAY – PINOUT & SPECIFICATIONS

Physical Appearance

- Small 0.96" (128×64 pixels) yellow/blue display
- 4-pin I2C header
- Integrated SSD1306 driver IC

Pinout Diagram

[OLED Module - Front View]



Pin Description

Pin #	Label	Function	Connection
1	GND	Ground	→ ESP32 GND
2	SCL	I2C Clock	→ ESP32 GPIO 22
3	SDA	I2C Data	→ ESP32 GPIO 21
4	VCC	Power (3.3V)	→ ESP32 3.3V

Technical Specifications

- **Display Type:** OLED (organic light-emitting diode)
- **Resolution:** 128 × 64 pixels
- **Display Size:** 0.96 inches diagonal
- **Color:** Blue background with yellow text
- **I2C Address:** 0x3C or 0x3D (default 0x3C)
- **Operating Voltage:** 3.3V (5V tolerant via onboard voltage divider)
- **Operating Current:** ~10–20 mA (varies with brightness)
- **Contrast:** User adjustable (0–255)

Wiring

ESP32		OLED
_____		_____
GPIO 21 (SDA)	————→	SDA (Pin 3)
GPIO 22 (SCL)	————→	SCL (Pin 2)
3.3V	————→	VCC (Pin 4)
GND	————→	GND (Pin 1)

Arduino Code Example (Using Adafruit Library)

```
#include <Adafruit_SSD1306.h>
#include <Wire.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_ADDR 0x3C

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

void setup() {
  Wire.begin(21, 22);  // SDA=21, SCL=22

  if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR)) {
    Serial.println("OLED init failed!");
    while(1);
  }

  display.setTextSize(1);
  display.setTextColor(SSD1306_WHITE);
  display.setCursor(0, 0);
  display.println("System Ready");
  display.display();
}

void loop() {
  display.clearDisplay();
  display.setCursor(0, 0);
  display.print("HR: 72 bpm");
  display.setCursor(0, 20);
  display.print("SpO2: 98%");
  display.display();
  delay(1000);
}
```

18650 LI-PO BATTERY – SPECIFICATIONS

Physical Specifications

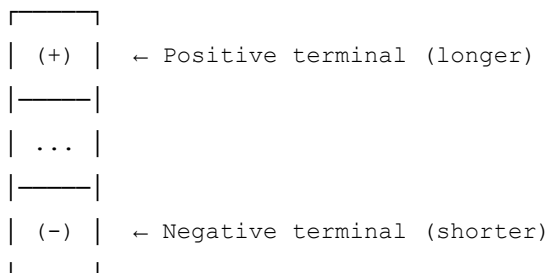
- **Type:** Lithium-ion rechargeable
- **Nominal Voltage:** 3.7V
- **Capacity:** 1000 mAh (your project spec)
- **Dimensions:** 18mm diameter × 65mm length
- **Weight:** ~45 grams
- **Lifespan:** ~500–1000 charge cycles

Electrical Specifications

Parameter	Value	Notes
Nominal Voltage	3.7V	Single cell
Fully Charged	4.2V	(TP4056 regulation)
Discharged	3.0V	Minimum safe level
Capacity	1000 mAh	~1 Ampere-hour
Typical Runtime	10–12 hours	With 100mA load
Max Discharge Rate	10A peak	Check battery rating
Internal Resistance	~50–100 mΩ	Varies with age

Pinout

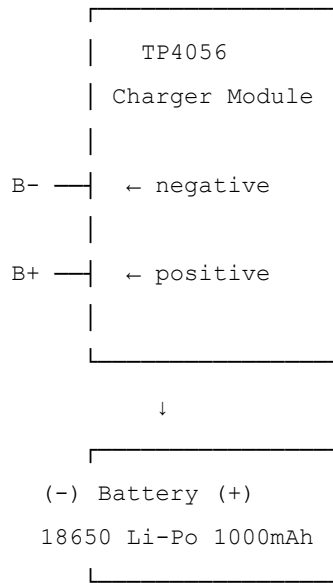
[18650 Battery - End View]



Protection Circuits

- **Over-voltage:** TP4056 prevents charging above 4.2V
- **Under-voltage:** Never discharge below 3.0V (prevents permanent damage)
- **Over-current:** Check battery specs (typically 10A)

Connection to TP4056



Safety Precautions

- **Never short** the terminals
- **Don't** expose to high heat ($>60^{\circ}\text{C}$)
- **Use** protective case/holder
- **Monitor** for swelling or leaks
- **Charge** only with TP4056 (never direct wall power)

PASSIVE COMPONENTS & MISCELLANEOUS

Resistors

Value	Quantity	Purpose	Notes
1 k Ω	3	Voltage divider (SIM800L RX)	1/4W carbon film
10 k Ω	5	Pull-up/pull-down resistors	1/4W carbon film
100 k Ω	2	Sensor biasing	1/4W carbon film
100 Ω	3	LED current limiting (UV LED)	1/4W carbon film
10 M Ω	1	High-impedance input (photodiode)	For transimpedance amp

Capacitors

Value	Quantity	Purpose	Notes
100 nF	10	Decoupling (power pins)	0.1 μF ceramic

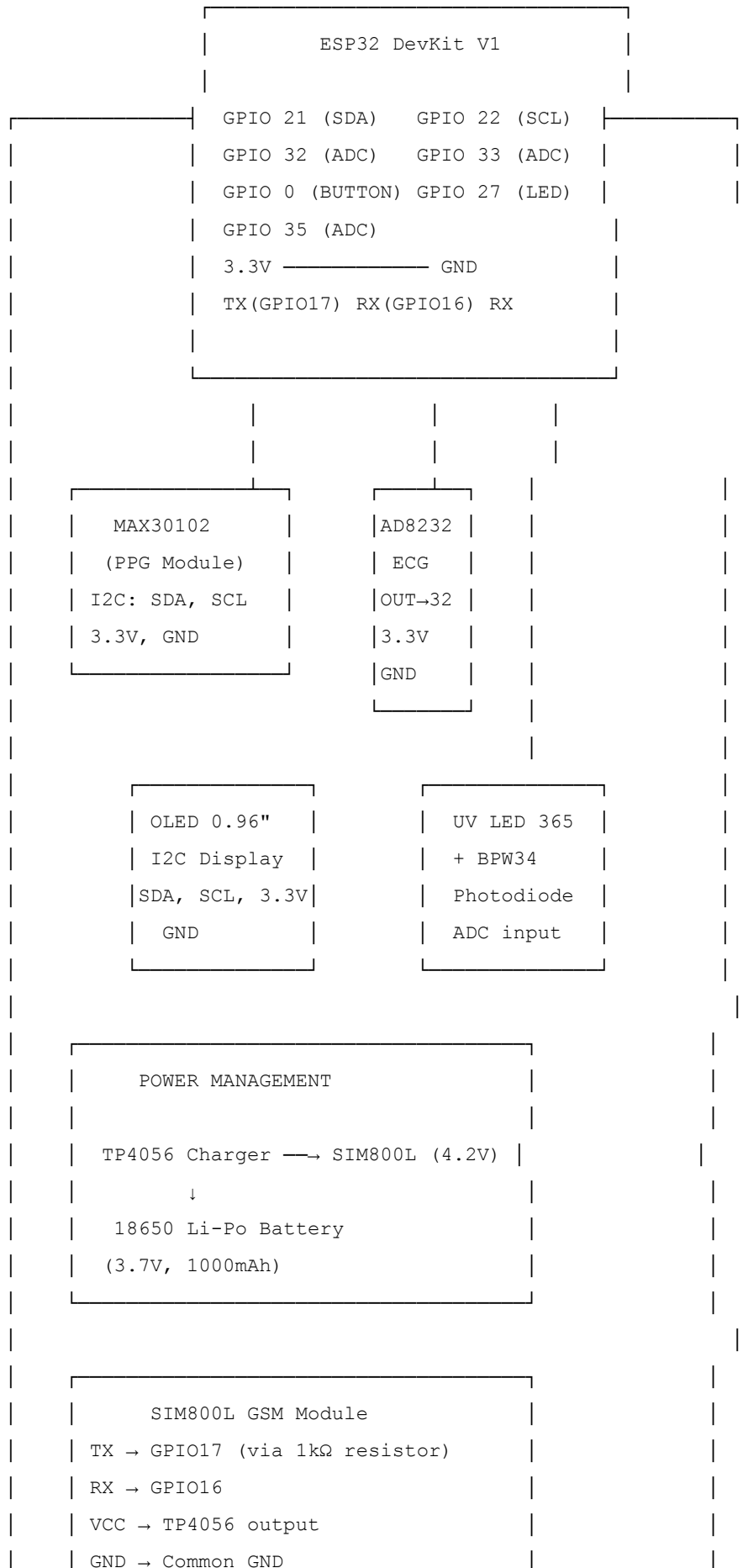
Value	Quantity	Purpose	Notes
1 μ F	5	General-purpose filtering	Ceramic or electrolytic
10 μ F	3	Power supply smoothing	Electrolytic
100 μ F	2	Large bulk capacitance	Electrolytic, 25V+
1000 μ F	1	SIM800L VCC buffering	Electrolytic, 6V+

Other Components

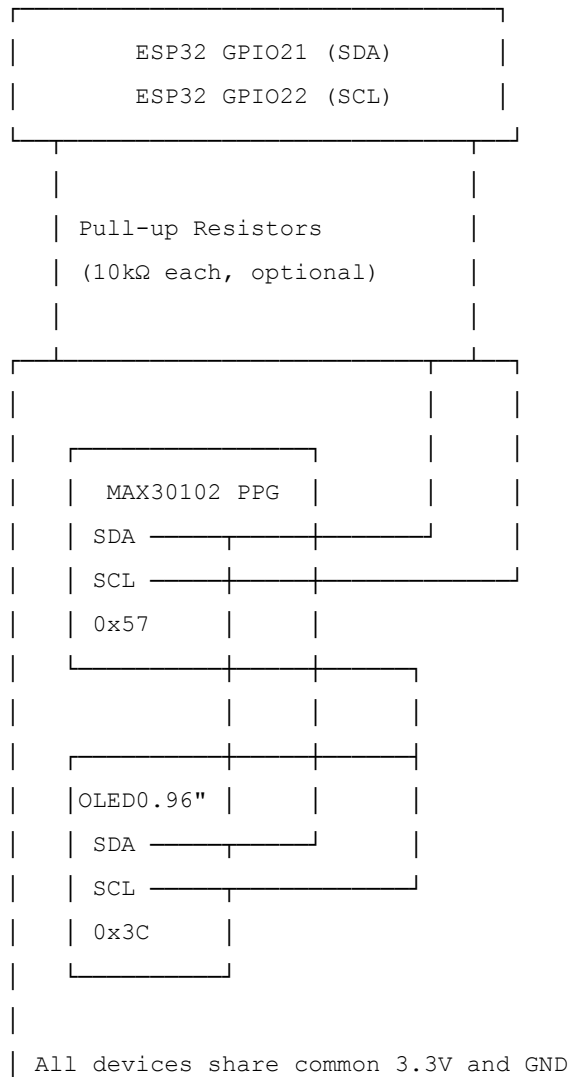
Component	Quantity	Purpose
2N7000 MOSFET	2	Digital switching (LED, relay control)
Push Buttons	3	User input (mode, reset, alert)
LED Indicators	5	Status indication (power, charging, alert)
Jumper Wires	1 pack	All interconnections
Breadboard	1	Prototype assembly
Conductive Pads/Electrodes	10	ECG electrode interface
Medical Silicone Patch	1 sheet	Biocompatible sensor housing

COMPLETE WIRING DIAGRAM

Master Wiring Schematic



I2C Bus Connection (Detailed)



I2C DEVICE ADDRESSES

Device	I2C Address (7-bit)	I2C Address (8-bit)	Notes
MAX30102	0x57	0xAE / 0xAF	Pulse oximeter
OLED SSD1306	0x3C	0x78 / 0x79	Default for most modules
(Alternative OLED)	0x3D	0x7A / 0x7B	If address jumper changed

Arduino I2C Scanner Code


```

#include <Wire.h>

void setup() {
  Serial.begin(115200);
  Wire.begin(21, 22);  // SDA=21, SCL=22 for ESP32

  Serial.println("\nI2C Device Scanner");
  byte error, address;
  int nDevices = 0;

  for (address = 1; address < 127; address++) {
    Wire.beginTransmission(address);
    error = Wire.endTransmission();

    if (error == 0) {
      Serial.print("I2C device found at address 0x");
      if (address < 16) Serial.print("0");
      Serial.println(address, HEX);
      nDevices++;
    }
  }

  if (nDevices == 0)
    Serial.println("No I2C devices found");
}

void loop() {}

```

POWER DISTRIBUTION

Power Budget Analysis

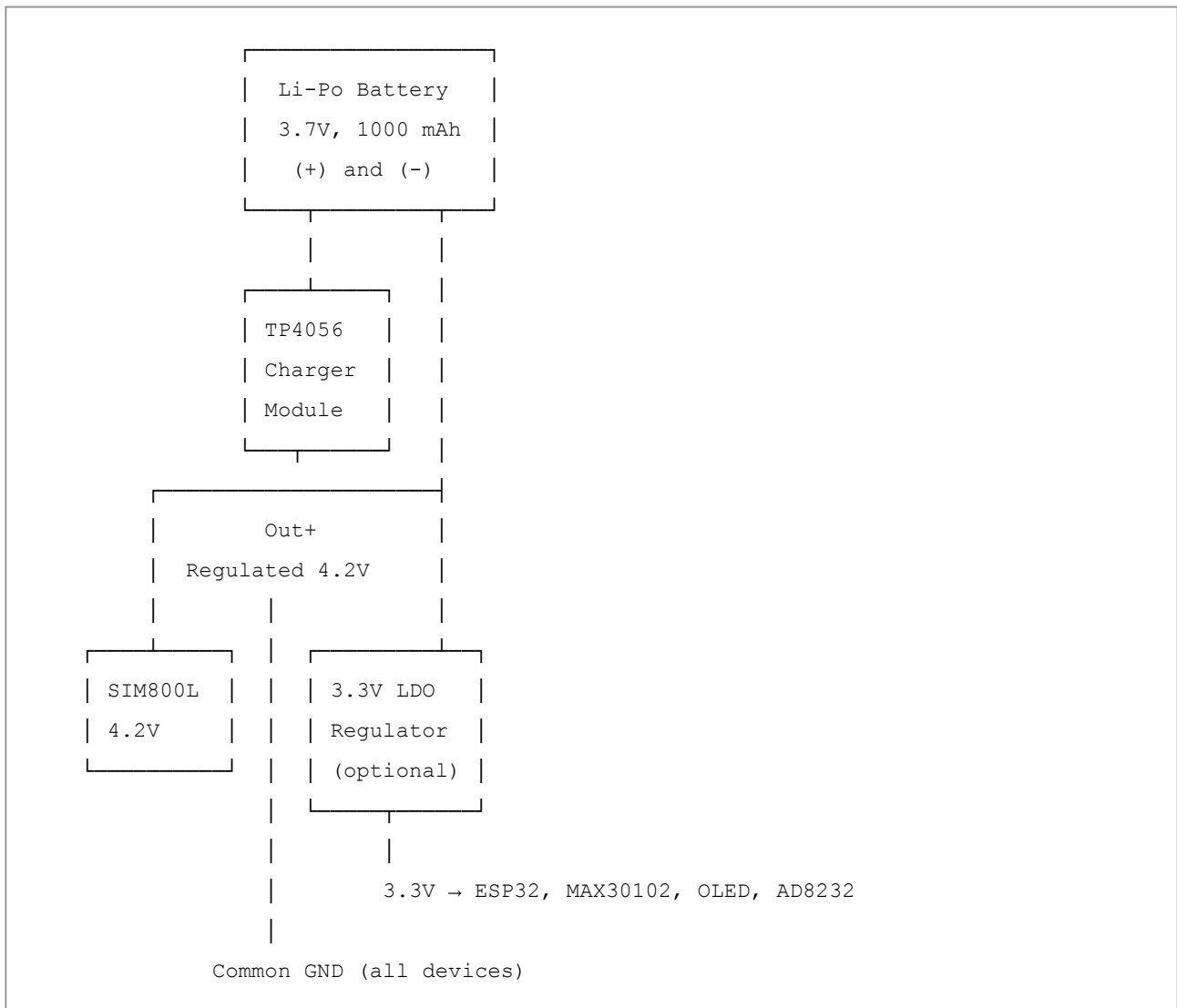
Device	Voltage	Current (Idle)	Current (Active)
ESP32	3.3V	10 mA	80–160 mA
MAX30102 PPG	3.3V	0.7 mA	100 mA (both LEDs)
AD8232 ECG	3.3V	5 mA	5 mA
OLED 0.96" Display	3.3V	5 mA	10–20 mA
SIM800L GSM	4.2V	5 mA	100–200 mA
UV LED 365nm	3.3V	0	20–30 mA
Total		~26 mA	300–500 mA

Battery Runtime Estimate

Using 18650 Li-Po 1000mAh battery:

- **Idle mode** (just running ESP32): ~40 hours
- **ECG + PPG active** (100 mA draw): ~10 hours
- **All sensors + GSM SMS** (300–500 mA peak): ~2–3 hours

Power Supply Connections



COMPONENTS SHOPPING CHECKLIST

Amazon.in Search Terms

- ☐ "ESP32 DevKit V1" or "ESP32-DEVKITC"
- ☐ "AD8232 ECG Module"

- ☐ "MAX30102 PPG Sensor Module"
- ☐ "OLED 0.96 inch I2C Display SSD1306"
- ☐ "SIM800L GSM Module Quad Band"
- ☐ "TP4056 Li-Po Charger Module"
- ☐ "18650 Battery 3.7V 1000mAh Rechargeable"
- ☐ "Jumper Wire Kit 40 Pin 20cm"
- ☐ "Resistor Assortment Kit 1/4W"
- ☐ "Capacitor Assortment Kit"
- ☐ "Micro USB Cable 1M"

Local Electronics Shops

- ☐ UV LED 365nm (5mm)
- ☐ BPW34 Photodiode
- ☐ 2N7000 MOSFET
- ☐ Push Buttons (6×6mm)
- ☐ LED Indicators (various colors)
- ☐ Medical-grade silicone patches
- ☐ Prototype breadboard

QUICK REFERENCE PIN TABLE

ESP32 Recommended Assignments

I2C Devices (MAX30102, OLED):

- └ SDA: GPIO 21
- └ SCL: GPIO 22

Analog Inputs (Biosensors):

- └ ECG (AD8232): GPIO 32
- └ Photodiode (BPW34): GPIO 35
- └ Battery Monitor: GPIO 33

Serial Communication (SIM800L):

- └ RX: GPIO 16
- └ TX: GPIO 17

Digital Outputs:

- └ UV LED Control: GPIO 27
- └ Status LED: GPIO 4
- └ Buzzer/Alert: GPIO 26

Digital Inputs:

- └ Push Button: GPIO 0

Power:

- └ 3.3V: Pin 1 (right side)
- └ GND: Pins 15, 16 (right side)
- └ GND: Pins 7, 1 (left side)

END OF COMPONENTS SPECIFICATION DOCUMENT

Total Pages: 15+ (when printed)

Last Updated: December 20, 2025

Status: Ready for Hardware Assembly

Next Step: Print this document and use it during hardware assembly and breadboard prototyping.